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EDUCATION

National Institute of Technology Jamshedpur

Aug 2023 – Present

B.Tech, Computer Science & Engineering — CGPA 8.52, Jamshedpur, Jharkhand

Interests: contrastive and self-supervised learning, domain adaptation, efficient attention and model compression, biomedical signal and image analysis, evaluation methodology.

RESEARCH

Nine papers: one published with a Best Paper Award, two accepted, six under review, five of them at SCIE-indexed journals (IEEE TBME, IEEE TMM, J. Neuroscience Methods, J. Franklin Institute, BSPC). First author on six.

Spectral-Separable Self-Attention for EEG Decoding

Published — IEEE DSPA 2026

Best Paper Award, 28th International DSPA Conference, Russia. First author. DOI

- Splits the signal into learnable frequency bands in the Fourier domain and attends within each band separately, which removes almost all of the attention cost of a standard transformer at 32K parameters.
- Works from three electrodes across three motor-imagery datasets, and the learned bands settle onto the mu, beta and gamma motor rhythms without being told to.

Cross-dataset contrastive pretraining for EEG decoding

Under review — IEEE TBME

First and corresponding author

- A large apparent gain from pretraining on five datasets disappeared under a matched control. The datasets were still separable by recording site after alignment, so the model had been learning which laboratory the data came from.
- Fixed with a label-free alignment step that adds no parameters and no inference cost, taking held-out accuracy from 67 to 75 per cent. Moving augmentation onto the GPU cut pretraining from around twenty hours to five.

Region-quality attention for lightweight NSFW detection

Under review — IEEE TMM

First author

- Feeds region quality into the attention gate rather than into the loss alone, so a 2.6M-parameter model scores above ViT and ConvNeXt baselines more than ten times its size.
- An audit of the benchmark found most evaluation positives duplicated from training images, and the two classes separable from file metadata alone. Every result was re-measured on a clean split.

Diversity-aware electrode selection for EEG

Under review — J. Neuroscience Methods

First and corresponding author

- Picks electrodes for complementary coverage rather than individual strength, keeping six of sixty-four without losing accuracy.
- Brings the model to a fifth of its parameters and inference to five milliseconds; the one comparison that did not reach significance is reported as such.

Learned electrode gating in a graph convolutional network

Under review — J. Franklin Institute

First and corresponding author

- Learns which electrodes matter during training instead of searching subsets afterwards, over a learned adjacency rather than one built from electrode distances.
- Drops 84 per cent of electrodes with no measurable accuracy cost, and beats two competing selection strategies at the same reduction.

Focal cortical dysplasia: a controlled benchmark and a false-positive stage

Under review — BSPC

Third author

- Four volumetric segmentation networks performed the same under one leakage-free protocol, which points to specificity rather than architecture as the limit at this data scale.
- A light re-scoring stage that attaches to any detector removed most false alarms while keeping nearly all lesions, bringing the burden from about seven per patient to under three.

InceptionWaveNet for epileptic seizure detection

Accepted — ICMVAI 2026, Springer LNEE

Second and corresponding author

- Replaces a fixed convolution block in EEGWaveNet with parallel branches and channel attention, improving cross-patient detection by eight points and holding up on an unseen cohort at 103K parameters.

Deep learning gradient boosting for remaining useful life prediction

Accepted — MECO 2026

Second author

- Four heterogeneous base learners corrected in sequence and combined through a stacking meta-learner on out-of-fold predictions, reaching an RMSE of 13.64 on the NASA C-MAPSS benchmark.

Hierarchical multi-scale fusion for remote sensing scene classification

Submitted — ICVGIP

Third author

- A top-down pathway spreads high-level meaning across every resolution of one backbone, and a small head fuses three pyramid levels into a single descriptor — multi-scale reasoning without an image pyramid or parallel streams.
- Improves on single-scale baselines across three backbones on AID and NWPU-RESISC45, with activation maps confirming that coarse and fine levels attend to different things.

EXPERIENCE

Machine Vision & Intelligence Lab — Research Intern

Dec 2025 – Present

NIT Jamshedpur

- Projects run end to end: data curation, training on a single consumer GPU, evaluation, manuscript writing and revision response.

ETS — AI Accessibility Summer Intern

May 2026 – Jul 2026

Remote, US

- Prototyped a face-centering accessibility solution for AI assessment interfaces, targeted for a US patent filing, and built an internal WCAG contrast-checker tool.
- Led an accessibility audit of eleven AI chatbot platforms against WCAG and EN 301 549, including conformance analysis of Anthropic, Google and Microsoft.

Flickerwall — Software Development Intern

Oct 2025 – Jan 2026

Remote, Bengaluru

- Multi-tenant subscription billing service in Go with Postgres and Redis, covering trial-to-active lifecycle, invoicing and proration, and Razorpay and Stripe integrations.
- On-premise signage server in Go with zone-based layouts, scheduling and licence validation, plus an admin dashboard in Next.js with Playwright end-to-end tests.

Official Web Team — Full Stack Developer

Dec 2024 – Present

NIT Jamshedpur — maintain the institute website serving 6.8M+ users, with 20+ production merges

PROJECTS

SafeSight — research model taken to a production service



PyTorch, ONNX, FastAPI, MLflow, Docker

- Serves the NSFW detector behind an API with a staged inference cascade that spends latency only on ambiguous images; the ONNX backend runs roughly three times faster than eager PyTorch on CPU.
- MLflow tracking and drift monitoring against a reference score window, with a build gate that fails the pipeline when recall regresses.

GSCC and NIT-JSR chatbot — retrieval-augmented information system



Node.js, Cohere, Pinecone, MongoDB, Redis

- Adapted for the Jharkhand Government's GSCC loan scheme, answering eligibility and application queries from official sources.
- A content-hash ledger re-indexes only changed documents, and semantic caching with conversation summarisation roughly halves tokens spent per request.

TECHNICAL SKILLS

Methods — contrastive and self-supervised pretraining, multiple-instance learning, graph neural networks, efficient attention, domain adaptation, model compression

Statistics — ROC comparison, bootstrap confidence intervals, paired significance testing, leakage and shortcut auditing

Tools — PyTorch, TensorFlow, MNE, ONNX, scikit-learn, LightGBM, XGBoost, NumPy, Pandas, MLflow, Docker, L^AT_EX

Languages — Python, C++, Java, Go, JavaScript, SQL

ACHIEVEMENTS

Best Paper Award — 28th International DSPA Conference, Russia (IEEE)

Amazon ML Summer School 2026 — selected among 1.34 lakh participants

Hackathons

HaXplore, IIT BHU — third of 1,100+ teams

Qualcomm Edge AI — top ten of 1,700+

Meta PyTorch OpenEnv — finalist of 31,000+

Peer reviewer — journal submissions in EEG and biomedical signal processing

Competitive programming — 1,500+ problems: LeetCode (Knight), CodeChef (1671), Codeforces (1404), HackerRank